

PubH7485 Homework2 Key

2024-09-27

Question 1 (20 points for each outcome, 60 points total)

The calculated results are summarized in the following tables:

Table 1: Outcome Table for Weight Change

	ATE	Standard Error	95% CI
Linear Regression	3.268	0.472	2.343, 4.193
Non-linear Term	3.364	0.476	2.432, 4.296
Interaction Term	3.33	0.451	2.446, 4.214

Table 2: Outcome Table for SBP

	ATE	Standard Error	95% CI
Linear Regression	1.356	0.936	-0.478, 3.189
Non-linear Term	1.441	1.02	-0.558, 3.44
Interaction Term	1.174	1.065	-0.914, 3.262

Table 3: Outcome Table for DBP

	ATE	Standard Error	95% CI
Linear Regression	1.314	0.666	0.009, 2.618
Non-linear Term	1.359	0.597	0.19, 2.529
Interaction Term	1.276	0.6	0.101, 2.452

Note that exact numbers may vary for students depending on the model fitting.

Bonus point (5 points for each outcome, 15 points total)

The calculated results are summarized in the following tables:

Table 4: Outcome Table by Stepwise Selection

	ATE	Standard Error	95% CI
Weight Change	3.202	0.428	2.364, 4.041
SBP	0.861	1.128	-1.35, 3.072
DBP	1.413	0.613	0.211, 2.614

The key is using stepwise selection but you can perform other variable selection methods like lasso, etc.

Appendix - R Code

1. Impute Missing Data

```
set.seed(7485)
data('nhefs')

# treatment
nhefs$qsmk
# outcomes of interest
nhefs$wt82_71
nhefs$sbp
nhefs$dbp

adj_vars <- c('sex', 'race', 'age', 'education', 'smokeintensity', 'smokeyrs', 'exercise', 'active', 'w

# imputation
cont_vars <- c('age', 'smokeintensity', 'smokeyrs', 'wt71', 'wt82_71', 'sbp', 'dbp')
cat_vars <- c('sex', 'race', 'education', 'exercise', 'active', 'hbp')

for (var in cont_vars) {
  nhefs[[var]][is.na(nhefs[[var]])] <- median(nhefs[[var]], na.rm = TRUE)
}

mode_function <- function(x) {
  x <- as.character(x)
  tbl <- table(x, useNA = "ifany")
  as.character(names(tbl)[which.max(tbl)])
}

for (var in cat_vars) {
  nhefs[[var]][is.na(nhefs[[var]])] <- mode_function(nhefs[[var]])
  nhefs[[var]] <- as.factor(nhefs[[var]])
}
```

2. Calculate ATE, standard error and 95% CI for each outcome

```
# function for calculating the ATE
calculate_ATE <- function(model_formula, data, num_bootstraps = 100) {

  wc_model <- lm(model_formula, data = data)

  nhefs_trt <- nhefs_con <- data
  nhefs_trt$qsmk = 1
  nhefs_con$qsmk = 0

  pred1_wc <- predict(wc_model, newdata = nhefs_trt, type = "response")
  pred0_wc <- predict(wc_model, newdata = nhefs_con, type = "response")
  ATE_wc <- mean(pred1_wc - pred0_wc)

  # Bootstrap replication for SE
  ate_wc_boot <- c()
  for (i in 1:num_bootstraps) {
```

```

nhefs_boot <- data[sample(nrow(data), size = nrow(data), replace = TRUE), ]
wc_model_boot <- lm(model_formula, data = nhefs_boot)

nhefs_trt_boot <- nhefs_con_boot <- nhefs_boot
nhefs_trt_boot$qsmk = 1
nhefs_con_boot$qsmk = 0

pred1_wc_boot <- predict(wc_model_boot, newdata = nhefs_trt_boot, type = "response")
pred0_wc_boot <- predict(wc_model_boot, newdata = nhefs_con_boot, type = "response")
ate_wc_boot <- c(ate_wc_boot, mean(pred1_wc_boot - pred0_wc_boot))
}

# Calculate standard error and confidence interval
SE <- sd(ate_wc_boot)
CI <- ATE_wc + c(-1, 1) * qnorm(0.975) * SE
CI_str <- paste0(round(CI[1], 3), ", ", round(CI[2], 3))

return(list(ATE = round(ATE_wc, 3), SE = round(SE, 3), CI = CI_str))
}

## for weight change
# linear regression
model_wc1 <- wt82_71 ~ qsmk + sex + race + age + education + smokeintensity + smokeyrs +
  exercise + active + wt71 + hbp
result_wc1 <- calculate_ATE(model_wc1, nhefs, num_bootstraps = 100)

print(result_wc1)

# add non-linear term
model_wc2 <- qsmk + sex + race + poly(age, 2) + education + poly(smokeintensity, 2) +
  poly(smokeyrs, 2) + exercise + active + poly(wt71, 2) + hbp
result_wc2 <- calculate_ATE(model_wc2, nhefs, num_bootstraps = 100)

print(result_wc2)

# add interaction term
model_wc3 <- wt82_71 ~ qsmk * (sex + race + age + education + smokeintensity + smokeyrs +
  exercise + active + wt71 + hbp)
result_wc3 <- calculate_ATE(model_wc3, nhefs, num_bootstraps = 100)

print(result_wc3)

## for sbp
# linear regression
model_sbp1 <- sbp ~ qsmk + sex + race + age + education + smokeintensity + smokeyrs +
  exercise + active + wt71 + hbp
result_sbp1 <- calculate_ATE(model_sbp1, nhefs, num_bootstraps = 100)
print(result_sbp1)

# add non-linear term
model_sbp2 <- sbp ~ qsmk + sex + race + poly(age, 2) + education + poly(smokeintensity, 2) +
  poly(smokeyrs, 2) + exercise + active + poly(wt71, 2) + hbp
result_sbp2 <- calculate_ATE(model_sbp2, nhefs, num_bootstraps = 100)

```

```

print(result_sbp2)

# add interaction term
model_sbp3 <- sbp ~ qsmk * (sex + race + age + education + smokeintensity + smokeyrs +
  exercise + active + wt71 + hbp)
result_sbp3 <- calculate_ATE(model_sbp3, nhefs, num_bootstraps = 100)
print(result_sbp3)

## for dbp
# linear regression
model_dbp1 <- dbp ~ qsmk + sex + race + age + education + smokeintensity + smokeyrs +
  exercise + active + wt71 + hbp
result_dbp1 <- calculate_ATE(model_dbp1, nhefs, num_bootstraps = 100)
print(result_dbp1)

# add non-linear term
model_dbp2 <- dbp ~ qsmk + sex + race + poly(age, 2) + education + poly(smokeintensity, 2) +
  poly(smokeyrs, 2) + exercise + active + poly(wt71, 2) + hbp
result_dbp2 <- calculate_ATE(model_dbp2, nhefs, num_bootstraps = 100)
print(result_dbp2)

# add interaction term
model_dbp3 <- dbp ~ qsmk * (sex + race + age + education + smokeintensity + smokeyrs +
  exercise + active + wt71 + hbp)
result_dbp3 <- calculate_ATE(model_dbp3, nhefs, num_bootstraps = 100)
print(result_dbp3)

```

3. Create summary table for each outcomes

```

rows <- c("ATE", "Standard Error", "95% CI")
columns <- c("Linear Regression", "Non-linear Term", "Interaction Term")

table_wc <- rbind(result_wc1, result_wc2, result_wc3)
colnames(table_wc) <- rows
rownames(table_wc) <- columns
kable(table_wc, caption = "Outcome Table for Weight Change")

table_sbp <- rbind(result_sbp1, result_sbp2, result_sbp3)
colnames(table_sbp) <- rows
rownames(table_sbp) <- columns
kable(table_sbp, caption = "Outcome Table for SBP")

table_dbp <- rbind(result_dbp1, result_dbp2, result_dbp3)
colnames(table_dbp) <- rows
rownames(table_dbp) <- columns
kable(table_dbp, caption = "Outcome Table for DBP")

```

4. Perform stepwise selection for each outcome (or you can perform lasso or other variable selection method)

```

## for weight change
# specify full model
full_wc <- lm (wt82_71 ~ qsmk * (sex + race + poly(age,2) + education + poly(smokeintensity,2) + poly(smokeyrs,2) +
exercise + active + poly(wt71,2) + hbp), data=nhefs)

# do stepwise selection
stepwise_wc <- stepAIC(full_wc, direction = "both")
summary(stepwise_wc)

select_wc <- wt82_71 ~ qsmk + sex + age + education + smokeyrs + active + wt71 + qsmk:education
result_wc4 <- calculate_ATE(select_wc, nhefs, num_bootstraps = 100)
print(result_wc4)

## for sbp
# specify full model
full_sbp <- lm (sbp ~ qsmk * (sex + race + poly(age,2) + education + poly(smokeintensity,2) + poly(smokeyrs,2) +
exercise + active + poly(wt71,2) + hbp), data=nhefs)

# do stepwise selection
stepwise_sbp <- stepAIC(full_sbp, direction = "both")
summary(stepwise_sbp)

select_sbp <- sbp ~ qsmk + sex + race + age + education + wt71 + hbp + qsmk:sex + qsmk:race + qsmk:hbp
result_sbp4 <- calculate_ATE(select_sbp, nhefs, num_bootstraps = 100)
print(result_sbp4)

## for dbp
# specify full model
full_dbp <- lm (dbp ~ qsmk * (sex + race + poly(age,2) + education + poly(smokeintensity,2) + poly(smokeyrs,2) +
exercise + active + poly(wt71,2) + hbp), data=nhefs)

# do stepwise selection
stepwise_dbp <- stepAIC(full_dbp, direction = "both")
summary(stepwise_dbp)

select_dbp <- dbp ~ qsmk + sex + race + smokeyrs + exercise + active + wt71 + hbp + qsmk : active
result_dbp4 <- calculate_ATE(select_dbp, nhefs, num_bootstraps = 100)
print(result_dbp4)

rows_bonus <- c("ATE", "Standard Error", "95% CI")
columns_bonus <- c("Weight Change", "SBP", "DBP")

table_bonus <- rbind(result_wc4, result_sbp4, result_dbp4)
colnames(table_bonus) <- rows_bonus
rownames(table_bonus) <- columns_bonus

kable(table_bonus, caption = "Outcome Table by Stepwise Selection")

```